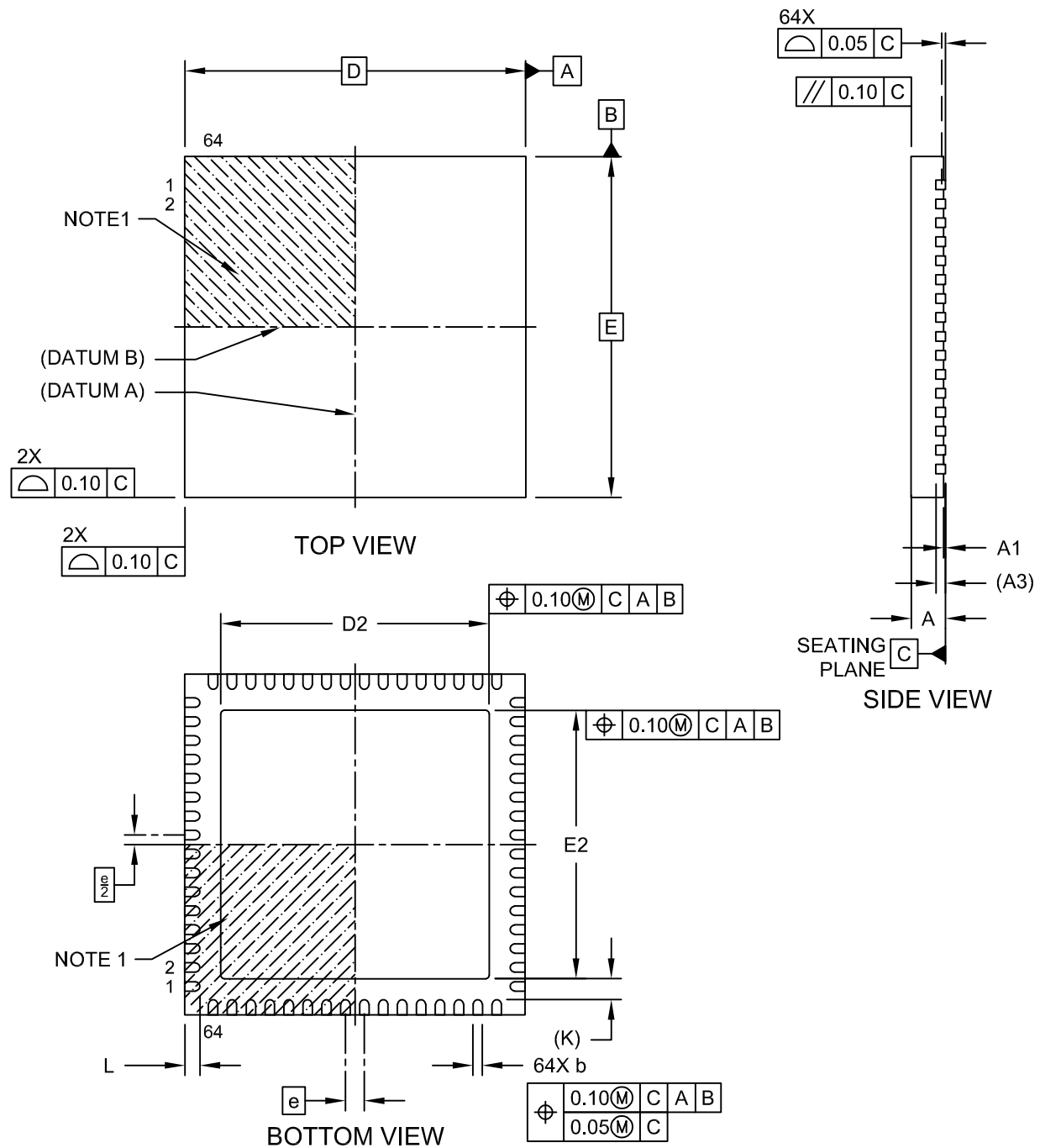


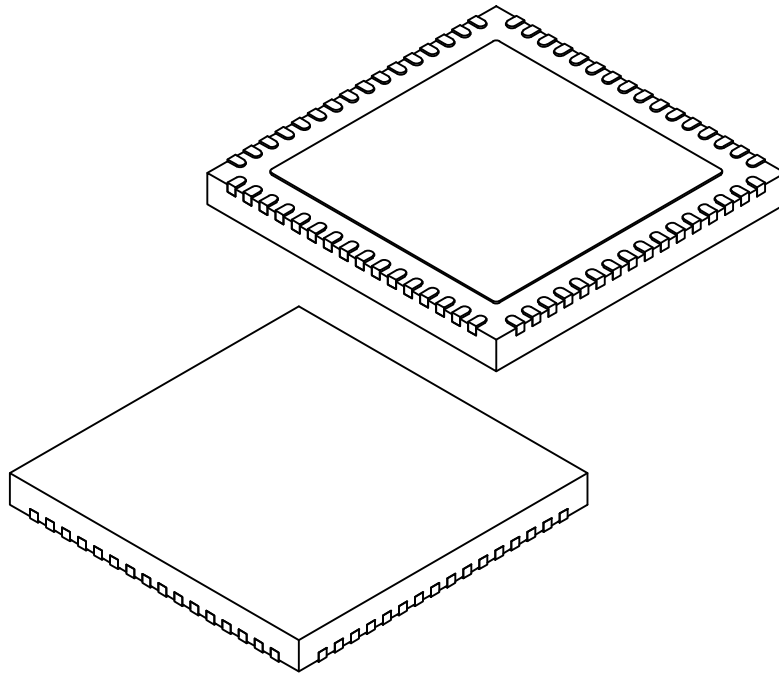
**64-Lead Very Thin Plastic Quad Flat, No Lead Package (THB) - 9x9x0.9 mm Body [VQFN]  
With 7.1 mm Square Exposed Pad; Atmel Legacy GPC ZGJ**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**64-Lead Very Thin Plastic Quad Flat, No Lead Package (THB) - 9x9x0.9 mm Body [VQFN]  
With 7.1 mm Square Exposed Pad; Atmel Legacy GPC ZGJ**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



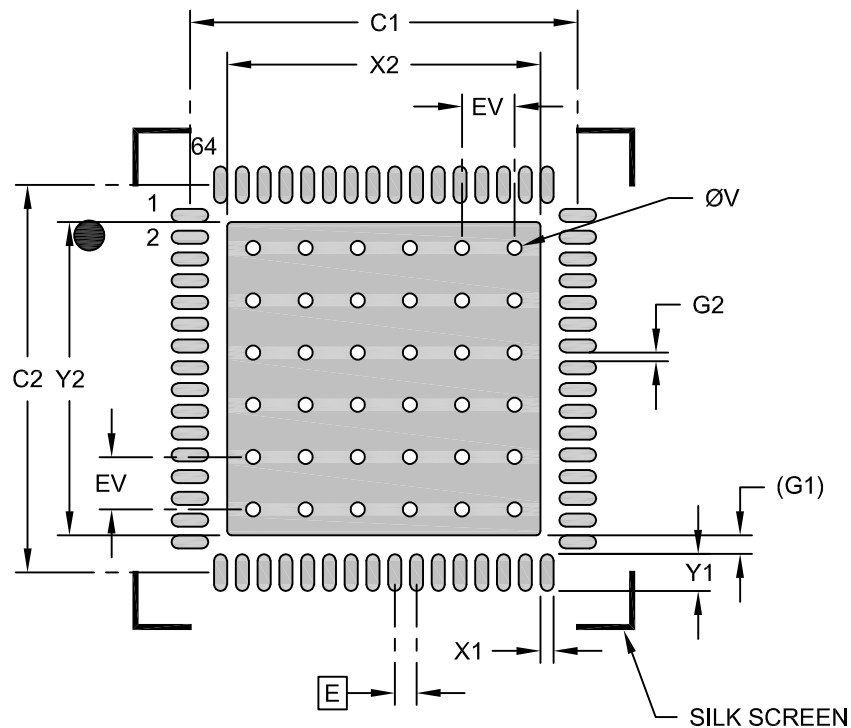
| Units                   |    | MILLIMETERS |          |      |
|-------------------------|----|-------------|----------|------|
| Dimension Limits        |    | MIN         | NOM      | MAX  |
| Number of Terminals     | N  | 64          |          |      |
| Pitch                   | e  | 0.50 BSC    |          |      |
| Overall Height          | A  | -           | -        | 0.90 |
| Standoff                | A1 | 0.00        | -        | 0.05 |
| Terminal Thickness      | A3 |             | 0.20 REF |      |
| Overall Length          | D  | 9.00 BSC    |          |      |
| Exposed Pad Length      | D2 | 6.95        | 7.10     | 7.25 |
| Overall Width           | E  | 9.00 BSC    |          |      |
| Exposed Pad Width       | E2 | 6.95        | 7.10     | 7.25 |
| Terminal Width          | b  | 0.23        | 0.25     | 0.28 |
| Terminal Length         | L  | 0.35        | 0.40     | 0.45 |
| Terminal-to-Exposed-Pad | K  | 0.35 REF    |          |      |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

**64-Lead Very Thin Plastic Quad Flat, No Lead Package (THB) - 9x9x0.9 mm Body [VQFN]  
With 7.1 mm Square Exposed Pad; Atmel Legacy GPC ZGJ**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**RECOMMENDED LAND PATTERN**

| Units                            |    | MILLIMETERS |      |      |
|----------------------------------|----|-------------|------|------|
| Dimension Limits                 |    | MIN         | NOM  | MAX  |
| Contact Pitch                    | E  | 0.50 BSC    |      |      |
| Optional Center Pad Width        | X2 |             |      | 7.25 |
| Optional Center Pad Length       | Y2 |             |      | 7.25 |
| Contact Pad Spacing              | C1 |             | 8.90 |      |
| Contact Pad Spacing              | C2 |             | 8.90 |      |
| Contact Pad Width (X64)          | X1 |             |      | 0.30 |
| Contact Pad Length (X64)         | Y1 |             |      | 0.85 |
| Contact Pad to Center Pad (X64)  | G1 | 0.43 REF    |      |      |
| Contact Pad to Contact Pad (X60) | G2 | 0.20        |      |      |
| Thermal Via Diameter             | V  |             | 0.33 |      |
| Thermal Via Pitch                | EV |             | 1.20 |      |

**Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23437 Rev A